Limits and Colimits in $\mathcal{R}\mathrm{ep}(G)$

Notes

Abstract

We explore a number of natural examples of limits and colimits in the category of representations of a finite group G. We shall see that limits correspond to subrepresentations and colimits correspond to quotient representations. We carefully distinguish these from the Hopf algebra multiplication and comultiplication on the group algebra $\mathbb{C}G$.

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1 Setup

Let G be a fixed finite group (for example S_n or $\mathrm{GL}_n(\mathbb{F}_q)$). A *representation* of G is a finite-dimensional complex vector space V together with a group homomorphism $\rho\colon G \to \mathrm{GL}(V)$. A morphism of representations $\varphi\colon V \to W$ is a $\mathbb{C}$-linear map satisfying $\varphi(g \cdot v) = g \cdot \varphi(v)$ for all $g \in G$, $v \in V$ (equivariance). Write $\mathcal{R}\mathrm{ep}(G)$ for the category of finite-dimensional complex representations of G.

2 Limits and colimits

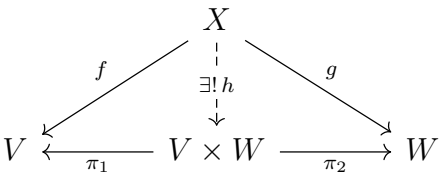
Start with some representations and some equivariant maps between them — call this a *diagram*.

Definition 2.1. The **limit** of a diagram is the *most general representation that maps into every object in the diagram, compatibly with all the maps*. "Most general" means: any other representation that maps in compatibly must factor through it uniquely.

The **colimit** is the dual: the *most general representation that every object in the diagram maps into*, compatibly.

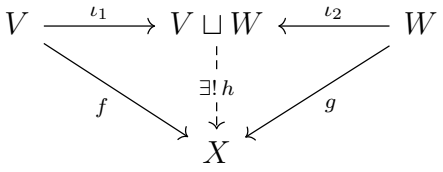
The simplest diagram is two representations with no maps between them: just V and W sitting side by side. The limit and colimit of this diagram are the **product** and **coproduct**.

The **product** $V \times W$ is the limit: the most general representation that maps into both V and W. It comes with projections $\pi_1 \colon V \times W \to V$ and $\pi_2 \colon V \times W \to W$, and its universal property says: for any X with equivariant maps $f \colon X \to V$ and $g \colon X \to W$, there is a unique $h \colon X \to V \times W$ with $\pi_1 \circ h = f$ and $\pi_2 \circ h = g$.



The arrows go *out* of $V \times W$. The universal property says $V \times W$ is the "best" representation that maps out to both V and W.

The **coproduct** $V \sqcup W$ is the colimit: the most general representation that both V and W map into. It comes with inclusions $\iota_1 \colon V \to V \sqcup W$ and $\iota_2 \colon W \to V \sqcup W$, and its universal property says: for any X with equivariant maps $f \colon V \to X$ and $g \colon W \to X$, there is a unique $h \colon V \sqcup W \to X$ with $h \circ \iota_1 = f$ and $h \circ \iota_2 = g$.



The arrows go *into* $V \sqcup W$. The universal property says $V \sqcup W$ is the "best" representation that both V and W map into.

In $\mathcal{R}\mathrm{ep}(G)$, the product turns out to equal the coproduct, every limit is a **subrepresentation**, and every colimit is a **quotient representation**. The object that is simultaneously the product and coproduct is the biproduct $V \oplus W$. We construct it next.

3 The biproduct

Definition 3.1. Given representations V and W, the *biproduct* $V \oplus W$ is:

- **Underlying space:** $\{(v, w) : v \in V,\ w \in W\}$.

- G**-action:** $g \cdot (v, w) = (g \cdot v,\ g \cdot w)$.
- **Four structure maps:**

$$V \xrightarrow{\iota_1} V \oplus W \xrightarrow{\pi_1} V$$

$$W \xrightarrow[\iota_2]{} V \oplus W \xrightarrow[\pi_2]{} W$$

where $\iota_1(v) = (v, 0)$, $\iota_2(w) = (0, w)$, $\pi_1(v, w) = v$, $\pi_2(v, w) = w$.

The structure maps satisfy:

$$\pi_1 \circ \iota_1 = \mathrm{id}_V \qquad \pi_1 \circ \iota_2 = 0 \tag{1}$$

$$\pi_2 \circ \iota_1 = 0 \qquad \pi_2 \circ \iota_2 = \mathrm{id}_W \tag{2}$$

All four maps are equivariant.

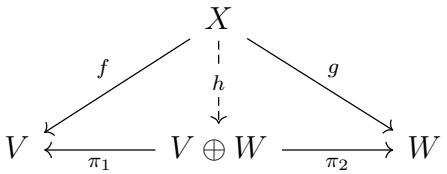
4 From biproduct to product

Forget the inclusions ι_1, ι_2. Keep only the projections.

The product is the limit of the diagram $V \quad W$ (two objects, no morphisms). A cone is an object X with maps $X \to V$ and $X \to W$; the universal cone is the product.

Proposition 4.1. *$(V \oplus W, \pi_1, \pi_2)$ is the product of V and W in $\mathcal{R}\mathrm{ep}(G)$.*

Proof. Given equivariant $f\colon X \to V$ and $g\colon X \to W$, we must find a unique equivariant $h\colon X \to V \oplus W$ with $\pi_1 \circ h = f$ and $\pi_2 \circ h = g$.



Existence. Define $h(x) = (f(x), g(x))$. Then $\pi_1 \circ h = f$ and $\pi_2 \circ h = g$. Equivariance: for any $\gamma \in G$, $h(\gamma \cdot x) = (f(\gamma \cdot x), g(\gamma \cdot x)) = (\gamma \cdot f(x), \gamma \cdot g(x)) = \gamma \cdot h(x)$.

Uniqueness. If $\pi_1 \circ h' = f$ and $\pi_2 \circ h' = g$, then

$$h'(x) = (\pi_1(h'(x)),\ \pi_2(h'(x))) = (f(x), g(x)) = h(x). \qquad \square$$

Remark 4.2. No addition of morphisms was needed. The product universal property only requires pairing, which is a set-level operation.

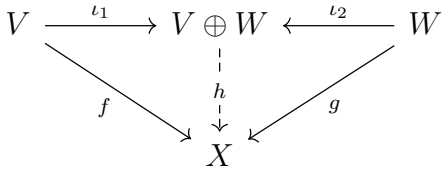
5 From biproduct to coproduct

Forget the projections π_1, π_2. Keep only the inclusions.

The coproduct is the colimit of the same diagram $V \quad W$. A cocone is an object X with maps $V \to X$ and $W \to X$; the universal cocone is the coproduct.

Proposition 5.1. *$(V \oplus W, \iota_1, \iota_2)$ is the coproduct of V and W in $\mathcal{R}\mathrm{ep}(G)$.*

Proof. Given equivariant $f\colon V \to X$ and $g\colon W \to X$, we must find a unique equivariant $h\colon V \oplus W \to X$ with $h \circ \iota_1 = f$ and $h \circ \iota_2 = g$.



Existence. Define $h(v, w) = f(v) + g(w)$. Then $h \circ \iota_1(v) = h(v, 0) = f(v) + g(0) = f(v)$, and similarly $h \circ \iota_2 = g$. Equivariance follows from linearity of f and g.

Uniqueness. Suppose $h' \circ \iota_1 = f$ and $h' \circ \iota_2 = g$. Then by linearity of h':

$$h'(v, w) = h'((v, 0) + (0, w)) = h'(\iota_1(v)) + h'(\iota_2(w)) = f(v) + g(w) = h(v, w). \qquad \square$$

Remark 5.2. This direction requires **addition**: we use $f(v) + g(w)$ to define h, and linearity of h' to prove uniqueness. This is why the coproduct universal property needs the additive structure of $\mathcal{V}ect$, while the product does not.

6 Limits: subspaces

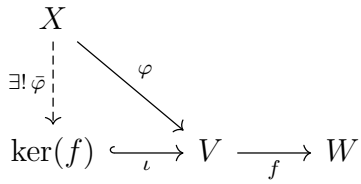
6.1 Kernel

Given equivariant $f\colon V \to W$, define $\ker(f) = \{v \in V : f(v) = 0\}$ with the inherited G-action, and let $\iota\colon \ker(f) \hookrightarrow V$ be the inclusion.

$$\ker(f) \xhookrightarrow{\iota} V \xrightarrow{f} W$$

First, $\ker(f)$ is a sub-representation: if $v \in \ker(f)$ and $g \in G$, then $f(g \cdot v) = g \cdot f(v) = g \cdot 0 = 0$, so $g \cdot v \in \ker(f)$.

Proposition 6.1 (Kernel is a limit)**.** $\ker(f)$ *satisfies the following universal property: for any equivariant* $\varphi\colon X \to V$ *with* $f \circ \varphi = 0$*, there exists a unique equivariant* $\bar{\varphi}\colon X \to \ker(f)$ *with* $\iota \circ \bar{\varphi} = \varphi$*.*



This is the universal property of a limit: $\ker(f)$ is the most general representation that maps into V with the constraint that the composite to W is zero. Any other such representation factors through it uniquely.

Proof. Since $f \circ \varphi = 0$, we have $\mathrm{im}(\varphi) \subseteq \ker(f)$, so $\bar{\varphi}(x) = \varphi(x)$ (viewed in $\ker(f)$) is well-defined and equivariant. Uniqueness: ι is injective, so $\iota \circ \bar{\varphi} = \varphi$ determines $\bar{\varphi}$ completely. $\square$

6.2 Pullback

The pullback is the limit of the cospan diagram $V \xrightarrow{f} X \xleftarrow{g} W$. A cone is an object Y with maps $Y \to V$ and $Y \to W$ making the square commute.

Concretely, given equivariant $f\colon V \to X$ and $g\colon W \to X$, the pullback is

$$V \times_X W = \{(v, w) \in V \oplus W : f(v) = g(w)\}$$

with the inherited G-action.

$$\begin{array}{ccc} V \times_X W & \xrightarrow{p_2} & W \\ {\scriptstyle p_1}\downarrow & \lrcorner & \downarrow{\scriptstyle g} \\ V & \xrightarrow[f]{} & X \end{array}$$

This is a subspace of $V \oplus W$ — precisely, it is the equaliser of $f \circ \pi_1$ and $g \circ \pi_2\colon V \oplus W \rightrightarrows X$.

7 Colimits: quotients

7.1 Cokernel

Given equivariant $f\colon V \to W$, define $\operatorname{coker}(f) = W/\operatorname{im}(f)$ with the quotient G-action $g \cdot [w] = [g \cdot w]$, and let $q\colon W \twoheadrightarrow \operatorname{coker}(f)$ be the projection.

$$V \xrightarrow{f} W \xrightarrow{q} \operatorname{coker}(f)$$

First, $\operatorname{im}(f)$ is a sub-representation: if $w = f(v) \in \operatorname{im}(f)$, then $g \cdot w = g \cdot f(v) = f(g \cdot v) \in \operatorname{im}(f)$. The quotient action is well-defined: if $w - w' \in \operatorname{im}(f)$, then $g \cdot w - g \cdot w' = g \cdot (w - w') \in \operatorname{im}(f)$, so $[g \cdot w] = [g \cdot w']$.

Proposition 7.1 (Cokernel is a colimit)**.** $\operatorname{coker}(f)$ *satisfies the following universal property: for any equivariant* $\psi\colon W \to X$ *with* $\psi \circ f = 0$*, there exists a unique equivariant* $\bar{\psi}\colon \operatorname{coker}(f) \to X$ *with* $\bar{\psi} \circ q = \psi$*.*

$$\begin{array}{ccccc} V & \xrightarrow{f} & W & \xrightarrow{q} & \operatorname{coker}(f) \\ & & & {\scriptstyle \psi}\searrow & \downarrow{\scriptstyle \exists!\,\bar{\psi}} \\ & & & & X \end{array}$$

This is the universal property of a colimit: $\operatorname{coker}(f)$ is the most general representation that W maps into with the constraint that the composite from V is zero. Any other such representation receives a unique map from it.

Proof. Since $\psi \circ f = 0$, we have $\operatorname{im}(f) \subseteq \ker(\psi)$, so $\bar{\psi}([w]) = \psi(w)$ is well-defined. Equivariance: $\bar{\psi}(g \cdot [w]) = \bar{\psi}([g \cdot w]) = \psi(g \cdot w) = g \cdot \psi(w) = g \cdot \bar{\psi}([w])$. Uniqueness: q is surjective, so $\bar{\psi} \circ q = \psi$ determines $\bar{\psi}$. □

7.2 Pushout

The pushout is the colimit of the span diagram $V \xleftarrow{f} X \xrightarrow{g} W$. A cocone is an object Y with maps $V \to Y$ and $W \to Y$ making the square commute.

Concretely, given equivariant $f\colon X \to V$ and $g\colon X \to W$, the pushout is

$$V \sqcup_X W = (V \oplus W)/\{(f(x), -g(x)) : x \in X\}$$

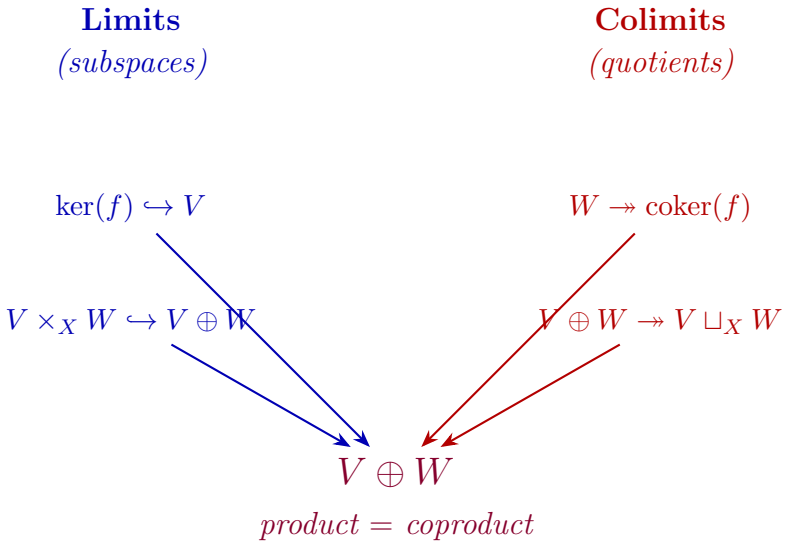
with the quotient G-action.

$$\begin{array}{ccc} X & \xrightarrow{\quad g \quad} & W \\ {\scriptstyle f}\downarrow & \ulcorner & \downarrow{\scriptstyle j_2} \\ V & \xrightarrow[\quad j_1 \quad]{} & V \sqcup_X W \end{array}$$

This is a quotient of $V \oplus W$ — precisely, it is the coequaliser of $\iota_1 \circ f$ and $\iota_2 \circ g\colon X \rightrightarrows V \oplus W$.

8 The big picture

Every limit in $\mathcal{R}\mathrm{ep}(G)$ is a **subspace** of something built from the inputs. Every colimit is a **quotient** of something built from the inputs. These two families meet at the biproduct:



The pullback is a subspace *of* the biproduct. The pushout is a quotient *of* the biproduct. So the biproduct is the common ground from which limits carve out subspaces and colimits form quotients.

9 Warning: the Hopf algebra is a different thing

The group algebra $\mathbb{C}G$ is a Hopf algebra, and this induces additional structure on the category $\mathcal{R}\mathrm{ep}(G)$. The Hopf algebra carries four operations, two of which are also sometimes called product and coproduct — a source of confusion.

To distinguish them from the categorical notions of the previous sections, we shall call them multiplication and comultiplication:

$$\begin{aligned}
\text{multiplication} \quad & \mu\colon \mathbb{C}G \otimes \mathbb{C}G \to \mathbb{C}G, & \mu(g_1 \otimes g_2) &= g_1 g_2,\\
\text{comultiplication} \quad & \Delta\colon \mathbb{C}G \to \mathbb{C}G \otimes \mathbb{C}G, & \Delta(g) &= g \otimes g,\\
\text{unit} \quad & \eta\colon \mathbb{C} \to \mathbb{C}G, & \eta(1) &= e,\\
\text{counit} \quad & \varepsilon\colon \mathbb{C}G \to \mathbb{C}, & \varepsilon(g) &= 1,\\
\text{antipode} \quad & S\colon \mathbb{C}G \to \mathbb{C}G, & S(g) &= g^{-1}.
\end{aligned}$$

These are all linear maps between vector spaces — they are morphisms in $\mathcal{V}ect$, not in $\mathcal{R}\mathrm{ep}(G)$. The Hopf algebra axioms are equations between these linear maps, diagrams that commute in $\mathcal{V}ect$. But they *induce* structure on $\mathcal{R}\mathrm{ep}(G)$:

- Δ gives the **tensor product**: if V and W are representations, then $V \otimes W$ becomes one via $g \cdot (v \otimes w) = gv \otimes gw$. The formula $gv \otimes gw$ comes from applying $\Delta(g) = g \otimes g$ to each tensor factor.
- ε gives the **unit**: the trivial representation $\mathbb{C}$ (where g acts as $\varepsilon(g) = 1$) satisfies $\mathbb{C} \otimes V \cong V \cong V \otimes \mathbb{C}$.
- S gives **duals**: the dual V^* has G-action $(g \cdot \varphi)(v) = \varphi(S(g) \cdot v) = \varphi(g^{-1} \cdot v)$.

So $\mathcal{R}\mathrm{ep}(G)$ has two independent structures. The biproduct $\oplus$ and the limits/colimits of sections 3–8 come from the *abelian* structure. The tensor product $\otimes$, the trivial representation, and duals come from the *Hopf algebra* structure on $\mathbb{C}G$. Together these make $\mathcal{R}\mathrm{ep}(G)$ into a *monoidal category*.

The theory of monoidal categories is beyond the scope of this document.

Remark 9.1. The limits and colimits of sections 3–8 are all computed by performing the construction in $\mathcal{V}ect$ — where the analysis is straightforward — and then checking that the G-action lifts to the result. This check is the content of every proof in sections 6–7: the kernel is a sub-representation because equivariance preserves the defining equation, the cokernel is a quotient representation because the image is G-stable, and so on. In this sense, the abelian structure of $\mathcal{R}\mathrm{ep}(G)$ is inherited from $\mathcal{V}ect$, but each construction requires verification that G respects it.

The Hopf algebra structure of this section is where G actually enters: the comultiplication $\Delta(g) = g \otimes g$ depends on G, and the monoidal structure it induces on $\mathcal{R}\mathrm{ep}(G)$ has no analogue for bare vector spaces.